ROSE 5710 HW1

Problem 1: Publisher Node

```python
#~/ros2_ws/src/hw1/publisher.py
import rclpy
from rclpy.node import Node
from std_msgs.msg import Float64MultiArray
import random
import math


class JointPublisher(Node):
    def __init__(self):
        super().__init__('joint_publisher')

        # Create publisher for joint angles with topic name including SID
        self.publisher_ = self.create_publisher(
            Float64MultiArray,
            '/hw1_q1_1155249290',
            10
        )

        # Create timer with 50Hz frequency (0.02 seconds period)
        self.timer = self.create_timer(0.02, self.timer_callback)

        self.get_logger().info(
            'Joint Publisher node started, publishing at 50Hz')

    def timer_callback(self):
        # Generate random joint angles (fix the random function call)
        joint_angles = [random.uniform(0, math.pi) for _ in range(6)]

        # Create Float64MultiArray message
        msg = Float64MultiArray()
        msg.data = joint_angles

        # Publish the joint angles
        self.publisher_.publish(msg)

        # Log the published data
        angles_str = [f"{angle:.3f}" for angle in joint_angles]
        self.get_logger().info(f'Published joint angles: {angles_str}')


def main(args=None):
    # Initialize ROS2
    rclpy.init(args=args)

    # Create the publisher node
    joint_publisher = JointPublisher()

    try:
        # Keep the node alive and responsive
```

```python
        rclpy.spin(joint_publisher)
    except KeyboardInterrupt:
        joint_publisher.get_logger().info('Publisher node stopped by user')
    finally:
        # Clean shutdown
        joint_publisher.destroy_node()
        rclpy.shutdown()


if __name__ == '__main__':
    main()

```

```bash
cd ~/ros2_ws

colcon build --packages-select hw1

source install/setup.bash

ros2 run hw1 publisher
```

```
(base) wzh@wzh-ASUS-TUF-Gaming-F16-FX607JV-FX607JV:~/Downloads/ros2_ws$ ros2 run hw1 publisher
[INFO] [1759146191.637482731] [joint_publisher]: Joint Publisher node started, publishing at 50Hz
[INFO] [1759146191.653054487] [joint_publisher]: Published joint angles: ['0.010', '0.155', '2.558', '2.378', '1.394', '1.044']
[INFO] [1759146191.673625355] [joint_publisher]: Published joint angles: ['3.120', '0.306', '0.495', '2.889', '1.263', '1.781']
[INFO] [1759146191.693533648] [joint_publisher]: Published joint angles: ['0.867', '0.640', '2.185', '1.765', '0.627', '2.511']
[INFO] [1759146191.713608258] [joint_publisher]: Published joint angles: ['2.361', '0.756', '2.735', '1.966', '0.806', '1.267']
[INFO] [1759146191.733749635] [joint_publisher]: Published joint angles: ['1.924', '2.486', '2.868', '2.367', '0.372', '1.393']
[INFO] [1759146191.753698470] [joint_publisher]: Published joint angles: ['2.535', '1.750', '0.770', '0.760', '2.981', '0.381']
[INFO] [1759146191.773094968] [joint_publisher]: Published joint angles: ['2.583', '0.517', '2.089', '0.120', '1.537', '1.834']
[INFO] [1759146191.793159991] [joint_publisher]: Published joint angles: ['1.014', '1.058', '2.621', '1.939', '2.851', '2.928']
[INFO] [1759146191.813137734] [joint_publisher]: Published joint angles: ['1.833', '2.770', '0.541', '1.627', '2.915', '0.557']
[INFO] [1759146191.833789821] [joint_publisher]: Published joint angles: ['1.404', '2.967', '1.345', '0.074', '2.769', '1.642']
[INFO] [1759146191.854756301] [joint_publisher]: Published joint angles: ['1.006', '1.842', '3.008', '2.005', '0.098', '0.536']
[INFO] [1759146191.873576946] [joint_publisher]: Published joint angles: ['2.167', '1.887', '1.767', '0.775', '1.849', '2.338']
[INFO] [1759146191.893446517] [joint_publisher]: Published joint angles: ['2.393', '0.888', '2.954', '1.627', '0.877', '0.340']
[INFO] [1759146191.913721035] [joint_publisher]: Published joint angles: ['1.164', '0.294', '1.570', '2.592', '1.013', '0.679']
[INFO] [1759146191.934519299] [joint_publisher]: Published joint angles: ['1.608', '0.631', '2.071', '2.135', '2.883', '1.789']
[INFO] [1759146191.953418586] [joint_publisher]: Published joint angles: ['0.501', '1.119', '1.623', '2.485', '1.313', '1.840']
[INFO] [1759146191.973399889] [joint_publisher]: Published joint angles: ['2.660', '1.145', '0.063', '1.001', '1.826', '3.016']
[INFO] [1759146191.993383472] [joint_publisher]: Published joint angles: ['3.099', '0.950', '2.944', '2.658', '0.230', '1.151']
[INFO] [1759146192.013241356] [joint_publisher]: Published joint angles: ['1.499', '1.698', '0.575', '2.336', '2.500', '3.041']
[INFO] [1759146192.033267541] [joint_publisher]: Published joint angles: ['2.414', '1.275', '2.282', '2.363', '1.256', '0.881']
[INFO] [1759146192.053762572] [joint_publisher]: Published joint angles: ['2.631', '1.072', '1.420', '1.124', '2.309', '1.412']
[INFO] [1759146192.073490611] [joint_publisher]: Published joint angles: ['2.045', '2.846', '0.517', '2.316', '0.736', '1.331']
```

```bash
ros2 topic echo /hw1_q1_1155249290
```

```
---
layout:
  dim: []
  data_offset: 0
data:
- 1.522544492160877
- 0.1263104387222467
- 3.121461996947035
- 1.8938184762653605
- 0.9462489153898428
- 0.9700185370535404
---
layout:
  dim: []
  data_offset: 0
data:
- 0.06356559656132268
- 3.1016788044054393
- 3.089817751415182
- 2.118335860440665
- 0.4038065806617453
- 1.9557953354399755
---
layout:
  dim: []
  data_offset: 0
data:
- 0.22392471512751208
- 0.32967990823435595
- 0.25227566574913524
- 1.8178546608028296
- 1.6325808533077548
- 0.28360763949632745
---
```

Problem 2: Subscriber + Publisher Node

```python
#~/ros2_ws/src/hw1/subscriber.py
import rclpy
from rclpy.node import Node
from std_msgs.msg import Float64MultiArray
import math
from geometry_msgs.msg import TransformStamped

# Import SOARM101 class
from hw1.SOARM101 import SOARM101


class FKSubscriber(Node):
    def __init__(self):
        super().__init__('fk_subscriber')
        # Initialize SOARM101
        self.arm = SOARM101()

        # Create subscriber for joint angles from Problem 1
        self.subscription = self.create_subscription(
            Float64MultiArray,
            '/hw1_q1_1155249290',
            self.listener_callback,
            10
        )

        # Create publisher for transformation data
        self.publisher_ = self.create_publisher(
            TransformStamped,
            '/hw1_q2_1155249290',
            10
        )
```

```python

        self.get_logger().info('FK Subscriber node started')

    def matrix_to_quaternion(self, rotation_matrix):
        """Convert rotation matrix to quaternion"""
        R = rotation_matrix
        trace = R[0, 0] + R[1, 1] + R[2, 2]

        if trace > 0:
            s = math.sqrt(trace + 1.0) * 2  # s=4*qw
            w = 0.25 * s
            x = (R[2, 1] - R[1, 2]) / s
            y = (R[0, 2] - R[2, 0]) / s
            z = (R[1, 0] - R[0, 1]) / s
        elif R[0, 0] > R[1, 1] and R[0, 0] > R[2, 2]:
            s = math.sqrt(1.0 + R[0, 0] - R[1, 1] - R[2, 2]) * 2  # s=4*qx
            w = (R[2, 1] - R[1, 2]) / s
            x = 0.25 * s
            y = (R[0, 1] + R[1, 0]) / s
            z = (R[0, 2] + R[2, 0]) / s
        elif R[1, 1] > R[2, 2]:
            s = math.sqrt(1.0 + R[1, 1] - R[0, 0] - R[2, 2]) * 2  # s=4*qy
            w = (R[0, 2] - R[2, 0]) / s
            x = (R[0, 1] + R[1, 0]) / s
            y = 0.25 * s
            z = (R[1, 2] + R[2, 1]) / s
        else:
            s = math.sqrt(1.0 + R[2, 2] - R[0, 0] - R[1, 1]) * 2  # s=4*qz
            w = (R[1, 0] - R[0, 1]) / s
            x = (R[0, 2] + R[2, 0]) / s
            y = (R[1, 2] + R[2, 1]) / s
            z = 0.25 * s

        return [x, y, z, w]

    def listener_callback(self, msg):
        joints = msg.data  # 6 joint angles in radians
        if len(joints) != 6:
            self.get_logger().warn(
                'Received incorrect number of joint angles!')
            return

        # Compute forward kinematics
        transform = self.arm.forward_kinematics(joints)

        if transform is None:
            self.get_logger().error('Forward kinematics failed!')
            return

        # Extract position (translation) from transformation matrix
        position = [
            round(float(transform[0, 3]), 4),
            round(float(transform[1, 3]), 4),
            round(float(transform[2, 3]), 4)
        ]

```

```python
        # Extract rotation matrix and convert to quaternion
        rotation_matrix = transform[:3, :3]
        quat = self.matrix_to_quaternion(rotation_matrix)
        quaternion = [round(q, 4) for q in quat]

        # Create TransformStamped message
        tf_msg = TransformStamped()
        tf_msg.header.stamp = self.get_clock().now().to_msg()
        tf_msg.header.frame_id = 'base_link'
        tf_msg.child_frame_id = 'end_effector'

        # Set translation
        tf_msg.transform.translation.x = position[0]
        tf_msg.transform.translation.y = position[1]
        tf_msg.transform.translation.z = position[2]

        # Set rotation (quaternion)
        tf_msg.transform.rotation.x = quaternion[0]
        tf_msg.transform.rotation.y = quaternion[1]
        tf_msg.transform.rotation.z = quaternion[2]
        tf_msg.transform.rotation.w = quaternion[3]

        # Publish the transformation
        self.publisher_.publish(tf_msg)

        self.get_logger().info(
            f'Published TF: pos=[{position[0]}, {position[1]}, '
            f'{position[2]}], quat=[{quaternion[0]}, {quaternion[1]}, '
            f'{quaternion[2]}, {quaternion[3]}]')


def main(args=None):
    # Initialize ROS2
    rclpy.init(args=args)

    # Create the subscriber node
    fk_subscriber = FKSubscriber()

    try:
        # Keep the node alive and responsive
        rclpy.spin(fk_subscriber)
    except KeyboardInterrupt:
        fk_subscriber.get_logger().info('FK Subscriber node stopped by user')
    finally:
        # Clean shutdown
        fk_subscriber.destroy_node()
        rclpy.shutdown()


if __name__ == '__main__':
    main()
```

```python
#~/ros2_ws/src/hw1/SOARM101.py
# DO NOT TOUCH!

import numpy as np


class SOARM101:
    """
    Forward Kinematics for SO Arm 101
    units: radians/cm
    use SOARM101() to initialize the class
    use arm.forward_kinematics(angles) to get the end effector
    transformation matrix
    """
    def __init__(self):
        # x, y, z, in local frames, in cm
        self.link_offsets = [
            [0, 3.115, 11.97],
            [0, 11.235, -2.8],
            [0, 13.49, 0.485],
            [0, 5.48, 0],
            [0, 3.15, 2],
            [0, 7.6, -1.15]
        ]

    def forward_kinematics(self, angles=None):
        """
        Forward Kinematics for SO Arm 101
        Args:
            angles: List of 6 joint angles in radians
        Returns:
            End effector transformation matrix, in cm
        """
        if angles is None:
            angles = [0, 0, 0, 0, 0, 0]

        if len(angles) != 6:
            return None

        # Initialize transformation matrix as identity
        T = np.eye(4)

        # For each joint, compute transformation matrix
        for i in range(6):
            # Get joint angle and link offset
            theta = angles[i]
            offset = self.link_offsets[i]

            # Create transformation matrix for this joint
            # Assuming Z-axis rotation joints (common for robotic arms)
            cos_theta = np.cos(theta)
            sin_theta = np.sin(theta)

            # Individual transformation matrix
            Ti = np.array([
```

```python
                [cos_theta, -sin_theta, 0, offset[0]],
                [sin_theta, cos_theta, 0, offset[1]],
                [0, 0, 1, offset[2]],
                [0, 0, 0, 1]
            ])

            # Accumulate transformations
            T = np.dot(T, Ti)

        return T


if __name__ == "__main__":
    arm = SOARM101()
    print(arm.forward_kinematics(angles=[0, 0, 0, 0, 0, 0]))

```

```bash
cd /home/wzh/Downloads/ros2_ws && colcon build --packages-select hw1

#bash 1
source install/setup.bash && ros2 run hw1 publisher

#bash 2
source install/setup.bash && ros2 run hw1 subscriber
```

- bash 1： publisher

```
(base) wzh@wzh-ASUS-TUF-Gaming-F16-FX607JV-FX607JV:~/Downloads/
ros2_ws$ source install/setup.bash && ros2 run hw1 publisher
[INFO] [1759147281.988987530] [joint_publisher]: Joint Publishe
r node started, publishing at 50Hz
[INFO] [1759147282.004813289] [joint_publisher]: Published join
t angles: ['0.383', '2.828', '0.053', '2.817', '2.097', '1.600'
]
[INFO] [1759147282.025224186] [joint_publisher]: Published join
t angles: ['2.709', '2.165', '1.979', '0.610', '1.938', '1.358'
]
[INFO] [1759147282.045642621] [joint_publisher]: Published join
t angles: ['0.384', '2.824', '1.375', '1.239', '2.409', '1.205'
]
[INFO] [1759147282.065371038] [joint_publisher]: Published join
t angles: ['2.421', '1.911', '1.233', '0.545', '2.683', '2.952'
]
[INFO] [1759147282.085061479] [joint_publisher]: Published join
t angles: ['0.747', '2.552', '2.289', '3.115', '1.812', '1.135'
]
[INFO] [1759147282.104838062] [joint_publisher]: Published join
t angles: ['2.472', '0.489', '1.433', '0.266', '1.666', '0.469'
]
[INFO] [1759147282.125139056] [joint_publisher]: Published join
t angles: ['0.591', '0.855', '1.130', '1.772', '1.638', '2.218'
]
[INFO] [1759147282.145414738] [joint_publisher]: Published join
t angles: ['1.910', '1.023', '2.079', '2.494', '2.692', '1.335'
]
[INFO] [1759147282.165670138] [joint_publisher]: Published join
t angles: ['2.649', '1.373', '1.947', '0.471', '1.780', '0.281'
]
[INFO] [1759147282.185265762] [joint_publisher]: Published join
t angles: ['2.129', '1.589', '1.441', '2.499', '2.105', '0.357'
]
[INFO] [1759147282.205631446] [joint_publisher]: Published join
t angles: ['2.622', '1.209', '0.667', '0.815', '2.031', '2.209'
]
[INFO] [1759147282.225276623] [joint_publisher]: Published join
t angles: ['0.530', '0.224', '1.101', '2.880', '1.421', '0.427'
]
[INFO] [1759147282.245539284] [joint_publisher]: Published join
t angles: ['2.893', '3.029', '1.322', '0.008', '1.554', '0.870'
]
[INFO] [1759147282.265419452] [joint_publisher]: Published join
t angles: ['0.931', '3.136', '1.107', '1.600', '2.618', '2.887'
]
[INFO] [1759147282.285261803] [joint_publisher]: Published join
```

- bash 2：subscriber

```
(base) wzh@wzh-ASUS-TUF-Gaming-F16-FX607JV-FX607JV:~/Downloads/
ros2_ws$ source install/setup.bash && ros2 run hw1 subscriber
[INFO] [1759147277.794517631] [fk_subscriber]: FK Subscriber no
de started
[INFO] [1759147282.386248504] [fk_subscriber]: Published TF: po
s=[-6.2377, -10.4298, 10.505], quat=[0.0, 0.0, -0.799, 0.6013]
[INFO] [1759147282.406589450] [fk_subscriber]: Published TF: po
s=[2.1624, 14.1773, 10.505], quat=[0.0, 0.0, 0.5246, 0.8514]
[INFO] [1759147282.426388631] [fk_subscriber]: Published TF: po
s=[-16.5421, 5.7759, 10.505], quat=[0.0, 0.0, 0.9896, 0.1435]
[INFO] [1759147282.446946009] [fk_subscriber]: Published TF: po
s=[5.808, 1.8028, 10.505], quat=[0.0, 0.0, 0.7876, 0.6162]
[INFO] [1759147282.466750670] [fk_subscriber]: Published TF: po
s=[-10.7067, -14.2043, 10.505], quat=[0.0, 0.0, 0.9767, -0.2147
]
[INFO] [1759147282.486312422] [fk_subscriber]: Published TF: po
s=[-18.5209, -16.0689, 10.505], quat=[0.0, 0.0, -0.5815, 0.8135
]
[INFO] [1759147282.506345517] [fk_subscriber]: Published TF: po
s=[-3.1256, -9.6173, 10.505], quat=[0.0, 0.0, 0.0065, 1.0]
[INFO] [1759147282.525477417] [fk_subscriber]: Published TF: po
s=[-21.8731, -3.072, 10.505], quat=[0.0, 0.0, 0.0958, 0.9954]
[INFO] [1759147282.546985086] [fk_subscriber]: Published TF: po
s=[-6.1661, 8.1265, 10.505], quat=[0.0, 0.0, -0.7606, 0.6492]
[INFO] [1759147282.566272867] [fk_subscriber]: Published TF: po
s=[-5.5741, 27.5859, 10.505], quat=[0.0, 0.0, 0.3383, 0.941]
[INFO] [1759147282.585916101] [fk_subscriber]: Published TF: po
s=[-20.5464, 18.1144, 10.505], quat=[0.0, 0.0, 0.9242, 0.382]
[INFO] [1759147282.605666733] [fk_subscriber]: Published TF: po
s=[-9.7028, -3.7257, 10.505], quat=[0.0, 0.0, 0.0181, 0.9998]
[INFO] [1759147282.625671737] [fk_subscriber]: Published TF: po
s=[-6.5851, -27.2524, 10.505], quat=[0.0, 0.0, -0.78, 0.6258]
[INFO] [1759147282.645923814] [fk_subscriber]: Published TF: po
s=[-28.4475, 16.8156, 10.505], quat=[0.0, 0.0, 0.9457, -0.325]
[INFO] [1759147282.666306006] [fk_subscriber]: Published TF: po
s=[2.9926, -12.1138, 10.505], quat=[0.0, 0.0, 0.9427, -0.3337]
[INFO] [1759147282.686090678] [fk_subscriber]: Published TF: po
s=[-10.1746, -24.7398, 10.505], quat=[0.0, 0.0, 0.9814, -0.1922
]
[INFO] [1759147282.705686362] [fk_subscriber]: Published TF: po
s=[-18.8081, -9.348, 10.505], quat=[0.0, 0.0, 0.869, 0.4949]
[INFO] [1759147282.725528427] [fk_subscriber]: Published TF: po
s=[-0.2594, -9.3376, 10.505], quat=[0.0, 0.0, 0.9888, -0.1495]
[INFO] [1759147282.746656233] [fk_subscriber]: Published TF: po
s=[-21.6775, 12.4889, 10.505], quat=[0.0, 0.0, -0.8507, 0.5257]
[INFO] [1759147282.766560690] [fk_subscriber]: Published TF: po
s=[-11.716, -8.8073, 10.505], quat=[0.0, 0.0, -0.1896, 0.9819]
[INFO] [1759147282.786283865] [fk_subscriber]: Published TF: po
s=[-9.1268, 26.2079, 10.505], quat=[0.0, 0.0, 0.996, 0.0895]
```

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